

Logistic regression (binomial GLM)

Inference labs use the five-step template: Hypothesis → Visualise → Assumptions → Conduct → Conclude.

Learning objectives

- Fit a logistic regression with `glm(family = binomial)` and translate coefficients into odds ratios.
- Construct a confusion matrix at a chosen probability threshold.
- Report the model with an odds ratio, an interval, and a brief note on discrimination.

Prerequisites

Linear regression material.

Background

Logistic regression models the log-odds of a binary outcome as a linear combination of predictors. The exponent of a coefficient is the odds ratio for a one-unit increase in the corresponding predictor. Because the relationship is linear on the log-odds scale and not the probability scale, a slope of 0.5 log-odds is a larger change in probability near 0.5 than near 0.95.

The model is fit by maximum likelihood with iteratively reweighted least squares. Standard errors are computed from the observed information, and Wald intervals on the log-odds scale are exponentiated to give intervals on the odds ratio. Profile-likelihood intervals are more trustworthy when the sample is small or the predictor is strongly associated with the outcome.

A confusion matrix at a single threshold tells only part of the story; the thresholds used in the clinic may not be the one that maximises accuracy. A later lab covers ROC curves and calibration, which give a threshold-free view of the same model.

Setup

```
library(tidyverse)
library(broom)
library(MASS)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Hypothesis

Using `MASS::Pima.tr`, can plasma glucose (`glu`) predict diabetes status (`type`)?

2. Visualise

```
mutate(type = factor(type, levels = c("No", "Yes")))

ggplot(pima, aes(type, glu, fill = type)) +
  geom_boxplot(alpha = 0.6, colour = "grey30") +
```

```
labs(x = "Diabetes", y = "Plasma glucose") +
theme(legend.position = "none")
```

3. Assumptions

Binary outcome, independent observations, log-odds linear in the predictors (or handled explicitly with a smooth/transformation).

```
# residuals are less informative in GLMs; check for linearity on the logit scale
pima |>
  mutate(p = fitted(fit),
         logit = qlogis(p)) |>
  ggplot(aes(glu, logit)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "loess", se = FALSE, colour = "steelblue") +
  labs(x = "Plasma glucose", y = "Fitted logit")
```

4. Conduct

```
glance(fit)
```

```
pred_class <- ifelse(pred_prob > 0.5, "Yes", "No")
table(pred_class, truth = pima$type)
```

5. Concluding statement

In `nrow(pima)` women, each 10 mg/dL increase in plasma glucose was associated with an odds ratio for diabetes of `round(exp(10 * coef(fit)["glu"]), 2)` (95% CI from profile: `round(exp(10 * confint(fit)["glu", 1]), 2)` to `round(exp(10 * confint(fit)["glu", 2]), 2)`). At a 0.5 threshold the classifier had `round(100 * mean(pred_class == pima$type), 1)%` accuracy; a threshold-free assessment appears in a later lab.

Common pitfalls

- Reporting the coefficient on the log-odds scale without exponentiating.
- Confusing probability and odds.
- Evaluating the model with accuracy at one threshold and nothing else.

Further reading

- Hosmer DW, Lemeshow S. *Applied Logistic Regression*.
- Harrell FE. *Regression Modeling Strategies*, ch. 10.
- Gelman A, Hill J. *Data Analysis Using Regression and Multilevel...*

Session info

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)